

Topology

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A Self Study

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Definition 1.1 (Metric Space). Let X be a non empty set, and $d : X \times X \rightarrow \mathbb{R}$ be a metric on X such that $\forall x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) + d(z, y) \geq d(x, y)$

Then the pair (X, d) is called a *metric space*.

Definition 1.2 (Inner Product Space). If V is a real vector space, and $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called an inner product if $\forall x, y, z \in V$:

1. $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0 \iff x = 0$
2. $\langle x, y \rangle = \langle y, x \rangle$
3. $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$
4. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ where $\alpha \in \mathbb{R}$

Then $(V, \langle \cdot, \cdot \rangle)$ is called an *inner product space*.

Definition 1.3 (Norm Linear Space). Let V be a real vector space and $\|x\|$ be the norm of a vector $x \in V$, then $(V, \|\cdot\|)$ is a norm linear space if $\forall x, y \in V$:

1. $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$.
2. $\|ax\| = |a| \|x\|$ where $a \in \mathbb{R}$.
3. $\|x + y\| \leq \|x\| + \|y\|$.

Theorem 1.4 (Cauchy–Schwarz Inequality). Let V be an inner product space and $\|x\| = \sqrt{\langle x, x \rangle}$ be the norm of $x \in V$, then for $x, y \in V$ we have:

$$|\langle x, y \rangle| \geq \|x\| \|y\| \quad (1)$$

Proof. First, we assume $\|x\| = \|y\| = 1$, then consider $\langle x - y, x - y \rangle$:

$$\langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle - 2\langle x, y \rangle \quad (2)$$

$$= 2(1 - \langle x, y \rangle) \geq 0 \implies \langle x, y \rangle \leq 1 \quad (3)$$

Similarly, for $\langle x + y, x + y \rangle$ we get $\langle x, y \rangle \geq -1$. Hence,

$$|\langle x, y \rangle| \leq 1 = \|x\| \|y\| \quad (4)$$

Now, for the general non-zero case, let $u = x/\|x\|$ and $v = y/\|y\|$, then $\|u\| = \|v\| = 1$ and $\langle u, v \rangle \leq 1$, by the previous case.

$$\left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle \leq 1 \implies \langle x, y \rangle \leq \|x\| \|y\| \quad (5)$$

The case for $\|x\| = 0$ or $\|y\| = 0$ are trivial. $\square$

Definition 1.5. Let V be a real vector space and $x, y, z \in V$. Then z lies between x and y $\iff \exists t \in [0, 1]$ such that $z = tx + (1 - t)y$.

Theorem 1.6. Let $x, y \in V$, where V is an inner product space with the norm metric. Then $d(x, y) = d(x, z) + d(z, y) \iff z$ lies between x and y .

Proof. If z lies between x and y , then $z = tx + (1 - t)y$ for some $t \in [0, 1]$, by the triangle inequality,

$$d(x, z) + d(z, y) = \|x - z\| + \|z - y\| \quad (6)$$

$$= \|x - tx + (t - 1)y\| + \|tx + (1 - t)y - y\| \quad (7)$$

$$= \|(1 - t)x - (1 - t)y\| + \|tx - ty\| \quad (8)$$

$$= (1 - t)\|x - y\| + t\|x - y\| = \|x - y\| \quad (9)$$

$$= d(x, y) \quad (10)$$

If we assume $d(x, y) = d(x, z) + d(z, y)$, then $\|x - y\| = \|x - z\| + \|z - y\|$, which implies that $x - z$ and $y - z$ are linearly dependent and $x - z = k(y - z)$ where $k \geq 0 \implies z = \frac{1}{1+k}x + \frac{k}{1+k}y$, and let $t = \frac{1}{1+k}$. Since $t \in [0, 1]$, we have $z = tx + (1 - t)y$ i.e. z lies between x and y . $\square$

Lemma 1.7 (Young's Inequality). Let x, y be two non negative real numbers. Let $p > 1$ and q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (11)$$

Proof. We fix $y > 0$, and define the following function:

$$f(x) := \frac{x^p}{p} + \frac{y^q}{q} - xy \quad (12)$$

Applying the extrema test of single variable, we get that $f'(x_0) = 0$, and see that it holds for $x_0 = y^{1/(p-1)}$. Then we also see that $f''(y^{1/(p-1)}) > 0$, from which we can conclude that it is a minimum of the function on $(0, \infty)$, from which it follows that $f(x) \geq f(x_0)$, and here $f(x_0)$ happens to be zero. Therefore:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (13)$$

□

Lemma 1.8 (Hölder's Inequality). *Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and for $1 \leq p \leq \infty$, we define $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ and for $p \rightarrow \infty$, we have $\|x\|_\infty = \max\{|x_i| : 1 \leq i \leq n\}$. For $p > 1$, let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:*

$$\sum_i |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad \forall a, b \in \mathbb{K}^n \quad (14)$$

Proof. Let us take $x = \frac{|a_i|}{\|a\|_p}$ and $y = \frac{|b_i|}{\|b\|_q}$ and using Young's Inequality, we get:

$$\frac{|a_i| |b_i|}{\|a\|_p \|b\|_q} \leq \frac{1}{p} \left(\frac{|a_i|}{\|a\|_p} \right)^p + \frac{1}{q} \left(\frac{|b_i|}{\|b\|_q} \right)^q \quad (15)$$

Summing this over i from 1 to n , we get:

$$\frac{1}{\|a\|_p \|b\|_q} \sum_{i=1}^n |a_i| |b_i| \leq \frac{1}{p \|a\|_p^p} \sum_{i=1}^n |a_i|^p + \frac{1}{q \|b\|_q^q} \sum_{i=1}^n |b_i|^q \quad (16)$$

Simplifying this, we obtain:

$$\sum_{i=1}^n |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad (17)$$

□

Lemma 1.9 (Minkowski's Inequality). *Let $1 \leq p \leq \infty$, then we have Minkowski's inequality:*

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad \forall a, b \in \mathbb{K}^n \quad (18)$$

Proof. For $1 < p < \infty$, we observe that:

$$\|a + b\|_p = \left(\sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \quad (19)$$

$$\leq \left(\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} + \sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \right) \quad (20)$$

Now, we apply Hölder's inequality to both the terms, by taking $p = q(p - 1)$ and $1/q = (p - 1)/p$. Applying it to the left term:

$$\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} \leq \|a\|_p \left[\sum_{i=1}^n (|a_i + b_i|^{p-1})^q \right]^{1/q} \quad (21)$$

$$= \|a\|_p \left[\sum_{i=1}^n |a_i + b_i|^p \right]^{1/q} \quad (22)$$

$$= \|a\|_p \|a + b\|_p^{p/q} \quad (23)$$

Similarly for the right term, we have

$$\sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \leq \|b\|_p \|a + b\|_p^{p/q} \quad (24)$$

Using these two relations and simplifying, we finally obtain:

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad (25)$$

□

Definition 1.10 (Open and Closed Balls). Let (X, d) be a metric space, and $x \in X$. Then for some $r > 0$, we define the open ball $B(x, r)$ and closed ball $B[x, r]$ as follows:

$$B(x, r) := \{y \in X \mid d(x, y) < r\} \quad (26)$$

$$B[x, r] := \{y \in X \mid d(x, y) \leq r\} \quad (27)$$

Definition 1.11 (Line Segment). Let V be a real vector space and $x, y \in V$. We denote the line segment joining x and y by $[x, y]$, and we define to be $[x, y] := \{tx + (1 - t)y \mid 0 \leq t \leq 1\}$. If $A \subset V$, then A is said to be *convex* if for any $x, y \in V$, the line segment $[x, y] \subset A$.

Definition 1.12 (Open Set). For some $U \subset (X, d)$, the subset is said to be *open* if $\exists r > 0$ such that $B(x, r) \subset U \forall x \in U$.

Lemma 1.13. Let (X, d) be a metric space, $x \in X$ and $r > 0$. Then the open ball $B(x, r)$ is an open set.

Proof. Let $y \in B(x, r)$. Assuming that $\exists s > 0$ such that $B(y, s) \subset B(x, r)$, if $z \in B(y, s)$, we want to show that $d(z, x) < r$. Now,

$$d(z, x) \leq d(z, y) + d(y, x) \leq s + d(x, y) < r$$

Therefore, we have $d(z, x) < r$, and thus $B(y, s) \subset B(x, r)$. Since y is arbitrary, we have proved the open ball $B(x, r)$ is an open set. □

Definition 1.14 (Interval). A subset $I \subset \mathbb{R}$ is an interval if and only if $\forall x, y \in I$ and $\forall z$ such that $x < z < y$, it follows that $z \in I$.

Proposition 1.15. Any subset $I \subset \mathbb{R}$ is an interval if and only if it is of the form (a, b) , $[a, b]$, $(a, b]$, $[a, b)$, (a, ∞) , $(-\infty, b)$ or $\mathbb{R}$ where $a \leq b$.